

HU Collection

TMUA Mock Set F Paper 1

Worked Solutions

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Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	B	C	C	E	D	C	D	A	C	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	F	B	D	E	E	A	A	D	D	C

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer B.

Convert everything to base 2. $\log_2 5 = \log_2 3 \cdot \log_3 5 = ab$. Then

$$\log_{15} 20 = \frac{\log_2 20}{\log_2 15} = \frac{2 + \log_2 5}{\log_2 3 + \log_2 5} = \frac{2 + ab}{a + ab} = \frac{2 + ab}{a(1 + b)}.$$

Option A treats $\log_2 5$ as b , forgetting that b is a base-3 logarithm.

Question 2. Answer C.

$2^{2x+3} = 4^{x+3/2}$, which is $y = 4^x$ translated $\frac{3}{2}$ to the left: I works. Also $2^{2x+3} = 8 \cdot 4^x$: II works. III gives $4^{2x} = 16^x$, and IV gives $4^x + 3$; neither is right. For exponential curves a horizontal translation and a vertical stretch are the same transformation, which is why two options succeed.

Question 3. Answer C.

The sum of n terms is $3(2^n - 1)$. We need $2^n - 1 > 333\,333.\bar{3}$, so $2^n \geq 333\,335$. Since $2^{18} = 262\,144$ is too small and $2^{19} = 524\,288$ is enough, $n = 19$. Using $2^{10} \approx 10^3$ carelessly gives $2^{20} \approx 10^6$, off by one in the wrong direction.

Question 4. Answer E.

Write $1 + x + x^2 = (1 + x) + x^2$ and expand by the binomial theorem: $\sum_j \binom{5}{j} (1 + x)^{5-j} x^{2j}$. An x^3 term needs $2j \leq 3$. For $j = 0$, take the x^3 coefficient of $(1 + x)^5$, which is 10. For $j = 1$, take 5 times the x coefficient of $(1 + x)^4$, which is $5 \cdot 4 = 20$. The total is 30.

Question 5. Answer D.

$x^{100} = (x - 1)(x - 2)q(x) + ax + b$. Putting $x = 1$ and $x = 2$ gives $a + b = 1$ and $2a + b = 2^{100}$, so $a = 2^{100} - 1$ and $b = 2 - 2^{100}$. Then $r(0) = b = 2 - 2^{100}$. The divisor's roots do all the work; long division is hopeless here.

Question 6. Answer C.

The graph of $y = x^3 - 3x^2$ has a local maximum at $(0, 0)$ and a local minimum at $(2, -4)$. A horizontal line $y = k$ meets it in exactly two points only when it is tangent at a turning point, that is $k = 0$ (roots $0, 0, 3$) or $k = -4$ (roots $2, 2, -1$). For $-4 < k < 0$ there are three solutions, and otherwise one. So two values of k .

Question 7. Answer D.

With $s = \sin x$: $2 - 2s^2 > 3s$, that is $2s^2 + 3s - 2 < 0$, or $(2s - 1)(s + 2) < 0$. Since $s + 2 > 0$ always, this is $\sin x < \frac{1}{2}$. That fails only on $[30^\circ, 150^\circ]$, of measure 120° , so the solution set has measure $360^\circ - 120^\circ = 240^\circ$.

Question 8. Answer A.

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (3 - k)^2 - 2k = 7$ gives $k^2 - 8k + 2 = 0$, so $k = 4 \pm \sqrt{14}$. But the roots must be real: $(k - 3)^2 - 4k \geq 0$, that is $k \leq 1$ or $k \geq 9$. Now $4 - \sqrt{14} \approx 0.26$ qualifies and $4 + \sqrt{14} \approx 7.74$ does not; there the formula still gives 7, but for a pair of complex roots. So $k = 4 - \sqrt{14}$ only.

Question 9. Answer C.

The logarithm needs $x^2 - 3x > 0$, that is $x < 0$ or $x > 3$. Since the base $\frac{1}{2}$ is less than 1, $\log_{1/2}$ is decreasing, so the inequality reverses: $x^2 - 3x < (\frac{1}{2})^{-2} = 4$, giving $-1 < x < 4$. Intersecting with the domain gives $-1 < x < 0$ or $3 < x < 4$. Forgetting the domain gives A, and forgetting the reversal gives B.

Question 10. Answer B.

$\tan$ has period 180° , so $\tan 3x = \tan x$ requires $3x = x + 180^\circ n$, that is $x = 90^\circ n$: the candidates are $0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$. But at 90° and 270° both $\tan x$ and $\tan 3x$ are undefined, so these are not solutions. That leaves $0^\circ, 180^\circ$ and 360° : three solutions.

Question 11. Answer F.

Touching both axes in the first quadrant means centre (r, r) and radius r . Tangency to the line requires $\frac{|3r+4r-12|}{5} = r$, that is $|7r - 12| = 5r$, giving $r = 6$ or $r = 1$. Both work: the circle of radius 1 is the incircle of the triangle cut off by the line, and the circle of radius 6 lies beyond the line. The sum is 7.

Question 12. Answer B.

Let $BM = MC = m$ and $\theta = \angle AMB$, so $\angle AMC = 180^\circ - \theta$ and $\cos \angle AMC = -\cos \theta$. The cosine rule in the two halves gives $25 = 16 + m^2 - 8m \cos \theta$ and $49 = 16 + m^2 + 8m \cos \theta$. Adding eliminates θ : $74 = 32 + 2m^2$, so $m^2 = 21$ and $BC = 2m = 2\sqrt{21}$. Option A is m itself.

Question 13. Answer D.

The curve and line meet at $x = 0$ and $x = k$, and the enclosed area is $\left| \int_0^k (kx - x^2) dx \right| = \frac{|k|^3}{6}$. Setting this equal to 36 gives $|k| = 6$, so $k = 6$ or $k = -6$. The two regions are mirror images in the y -axis. Writing $\int_0^k$ without the modulus finds only $k = 6$.

Question 14. Answer E.

From $y(1) = -4$: $1 + p + q = -4$. From $y'(1) = 0$: $3 + 2p + q = 0$. Subtracting gives $p = 2$ and $q = -7$. Then $y' = 3x^2 + 4x - 7 = (3x + 7)(x - 1)$, so the other stationary point is at $x = -\frac{7}{3}$, where

$$y = -\frac{343}{27} + 2 \cdot \frac{49}{9} + \frac{49}{3} = \frac{-343+294+441}{27} = \frac{392}{27}.$$

Question 15. Answer E.

The parabola and line meet where $x^2 = 2x + 3$, at $x = -1$ and $x = 3$. With $x \geq 0$ the region runs from 0 to 3, and its area is $\int_0^3 (2x + 3 - x^2) dx = 9 + 9 - 9 = 9$. Option F is the whole segment from -1 to 3, before the condition $x \geq 0$ is applied.

Question 16. Answer A.

The tangent to $y = x^2$ at $x = a$ is $y = 2ax - a^2$. It touches $y = -(x-3)^2$ when $x^2 + (2a-6)x + 9 - a^2 = 0$ has a repeated root: $(2a-6)^2 - 4(9-a^2) = 8a^2 - 24a = 0$, so $a = 0$ or $a = 3$. The common tangents are $y = 0$ and $y = 6x - 9$, meeting at $(\frac{3}{2}, 0)$. By symmetry, the two parabolas are images of each other under a half-turn about $(\frac{3}{2}, 0)$, so the meeting point had to be there.

Question 17. Answer A.

If the cylinder has height h and radius ρ , then $\rho^2 + (\frac{h}{2})^2 = 9$, so $V = \pi \left(9 - \frac{h^2}{4}\right) h$. Then $V'(h) = \pi \left(9 - \frac{3h^2}{4}\right) = 0$ at $h = 2\sqrt{3}$, a maximum. This gives $V = \pi(9-3) \cdot 2\sqrt{3} = 12\sqrt{3}\pi$.

Question 18. Answer D.

Multiples of 3: $3, 6, \dots, 498$, which is 166 terms with sum $3 \cdot \frac{166 \cdot 167}{2} = 41\,583$. Multiples of 7: $7, \dots, 497$, which is 71 terms with sum $7 \cdot \frac{71 \cdot 72}{2} = 17\,892$. Multiples of 21: $21, \dots, 483$, which is 23 terms with sum $21 \cdot \frac{23 \cdot 24}{2} = 5\,796$. “Not both” removes the multiples of 21 from both lists: $41\,583 + 17\,892 - 2 \cdot 5\,796 = 47\,883$. Subtracting 5796 only once gives the inclusive “or”, option E.

Question 19. Answer D.

Put $u = \sqrt{x^2 + 2} \geq \sqrt{2}$. The expression is $\frac{u^2+1}{u} = u + \frac{1}{u}$. The inequality $u + \frac{1}{u} \geq 2$ is useless here: equality needs $u = 1$, which is not allowed. Since $u + \frac{1}{u}$ is increasing for $u \geq 1$, the minimum is at $u = \sqrt{2}$, that is $x = 0$, giving $\sqrt{2} + \frac{1}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$. Option C is the unattainable bound.

Question 20. Answer C.

$f'(x) = 3(x-1)(x-3)$, with a local maximum $f(1) = 4 + c$ and a local minimum $f(3) = c$. Three distinct real roots need $f(1) > 0 > f(3)$, that is $-4 < c < 0$. They are all positive exactly when the smallest root is: since $f(0) = c < 0$ and $f(1) > 0$, the smallest root lies in $(0, 1)$. So $c \in \{-3, -2, -1\}$, three integers. At $c = 0$ one root is 0, not positive, and at $c = -4$ a repeated root appears.